Year 9 Mathematics

Unit 2: Area and Volume

2.2 Volume of Prisms

Volume of Prisms

STARTER · 5 MIN

Date: _____ **Title: Volume of Prisms**

LEARNING OBJECTIVES

- Find the volume of cuboids and triangular prisms
- Find the volume of any prism
- Find a missing length from the volume

STARTER *Answer in your book. No calculators.*

1. Area of a triangle: base 8 cm, height 5 cm
2. Volume of a cuboid 5 cm by 4 cm by 3 cm
3. How many cm^3 in 1 litre?
4. Work out $360 \div 24$

Starter — Answers

STARTER

Tick or correct in green pen

1. **20 cm²**
2. **60 cm³**
3. **1000 cm³**
4. **15**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the volume of cuboids and triangular prisms	☐
SILVER	I can find the volume of any prism	☐
GOLD	I can find a missing length from the volume	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Volume of Prisms

INTRODUCTION · 3 MIN

The volume of a prism is the area of its cross-section times its length.

Formula

Find the area of the end first.

V = area of cross-section × length

Units

Volume is measured in cubic units.

cm^3, m^3

Capacity

1 cm^3 holds 1 ml.

1000 cm^3 = 1 litre

Missing length

Divide the volume by the area.

length = V ÷ area

Examples 1 and 2 — Cuboids and Triangular Prisms

EXAMPLES · 5 MIN

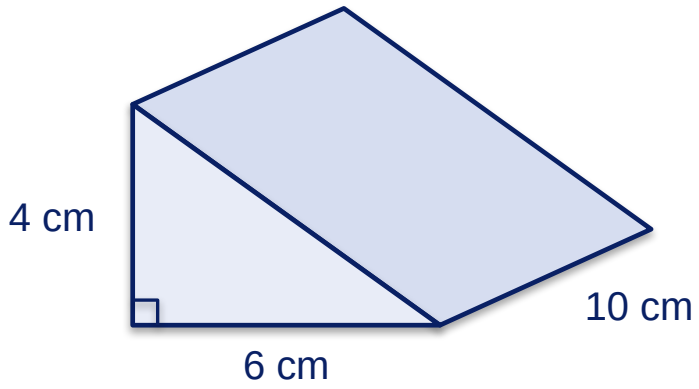
BRONZE

Watch and copy into your book

Hint V = area of cross-section × length 1000 cm^3 = 1 litre

Example 1

Find the volume.



Example 2

A fish tank is 50 cm by 30 cm by 40 cm. How many litres does it hold?

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Area of the triangle first

$$\text{area} = \frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$$

$$V = 12 \times 10$$

$= 120 \text{ cm}^3$

Tip: the triangle is the cross-section.

Example 2

Volume, then litres

$$50 \times 30 \times 40 = 60\,000 \text{ cm}^3$$

$$60\,000 \div 1000$$

$= 60$ litres

Tip: 1000 cm^3 = 1 litre

Task 1

PRACTICE · 9 MIN

BRONZE *Find the volume*

Hint V = area of cross-section × length 1000 cm³ = 1 litre

1. Cuboid 6 cm by 5 cm by 4 cm

2. Triangular prism: triangle base 8 cm, height 5 cm, length 10 cm

3. Triangular prism: triangle base 3 cm, height 4 cm, length 7 cm

4. A box is 20 cm by 10 cm by 10 cm. How many litres does it hold?

5. Cross-section 15 cm², length 8 cm

★ A tank is 40 cm by 25 cm by 30 cm. How many litres does it hold?

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **120 cm³**
2. **200 cm³**
3. **42 cm³**
4. **2000 cm³ = 2 litres**
5. **120 cm³**

★ **30 000 cm³ = 30 litres**

Examples 3 and 4 — Trapezium and Parallelogram Prisms

EXAMPLES · 5 MIN

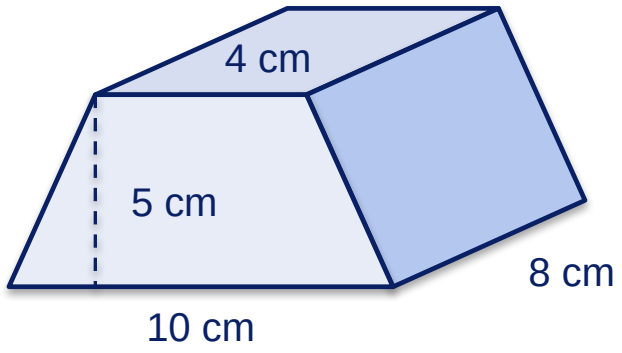
SILVER

Watch and copy into your book

Hint trapezium = $\frac{1}{2}(a + b)h$

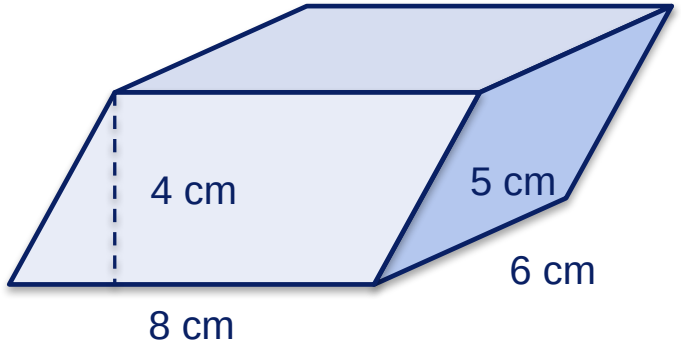
Example 3

Find the volume.



Example 4

Show that the volume is 192 cm³.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Area of the trapezium

$$A = \frac{1}{2} \times (4 + 10) \times 5 = 35 \text{ cm}^2$$

$$V = 35 \times 8$$

$= 280 \text{ cm}^3$

Tip: the height is 5 cm, not 8 cm.

Example 4

Parallelogram: base × height

$$\text{area} = 8 \times 4 = 32 \text{ cm}^2$$

$$V = 32 \times 6$$

$= 192 \text{ cm}^3$

Tip: 'show that': write every step.

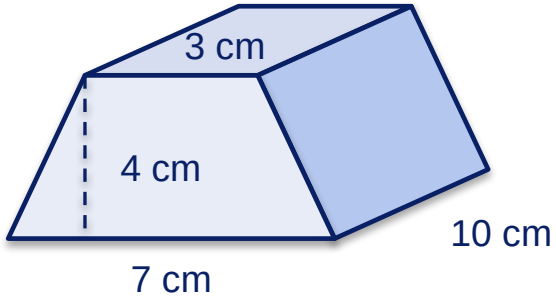
Task 2

PRACTICE · 9 MIN

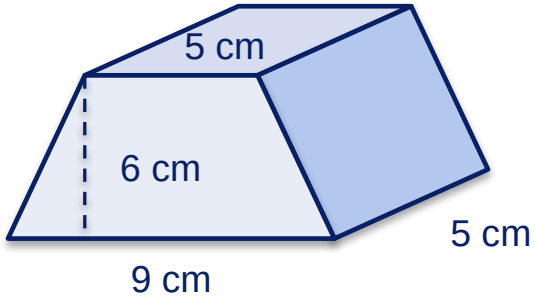
SILVER *Find the volume*

Hint trapezium = $\frac{1}{2}(a + b)h$

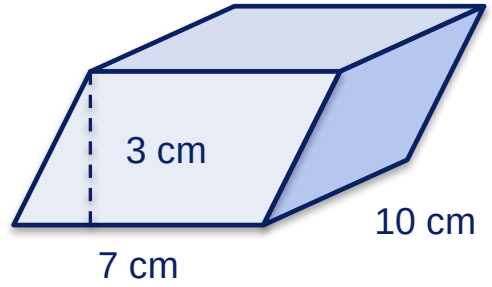
1 Volume



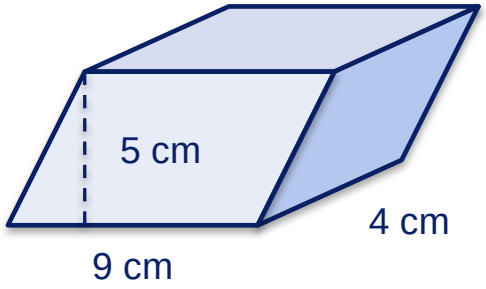
2 Volume



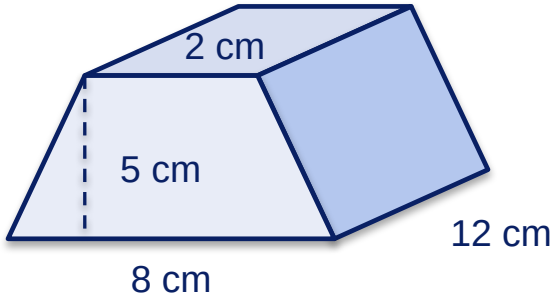
3 Volume



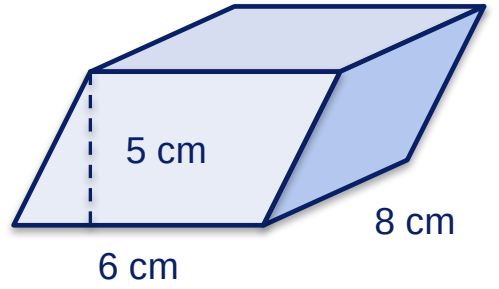
4 Volume



5 Volume



★ Show V = 240 cm³



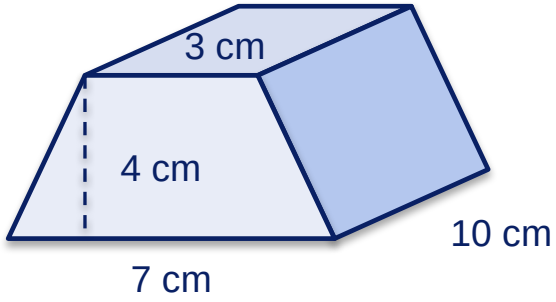
Task 2 — Answers

CORRECTIONS

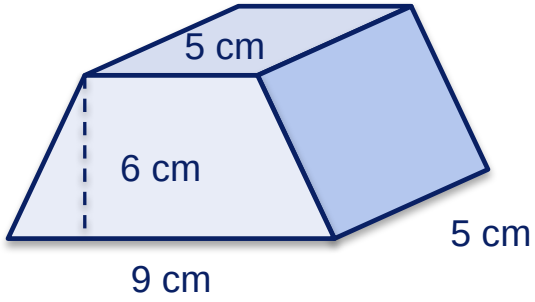
SILVER

Tick or correct in green pen

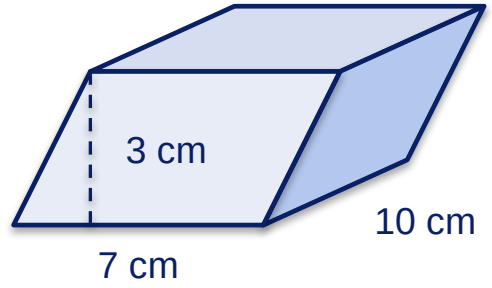
1 **200 cm³**



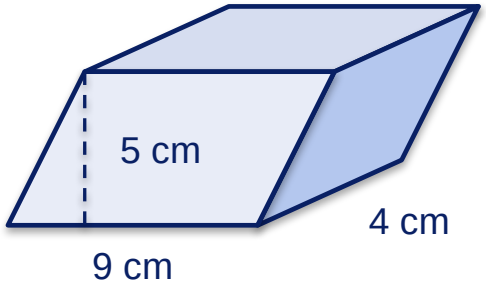
2 **210 cm³**



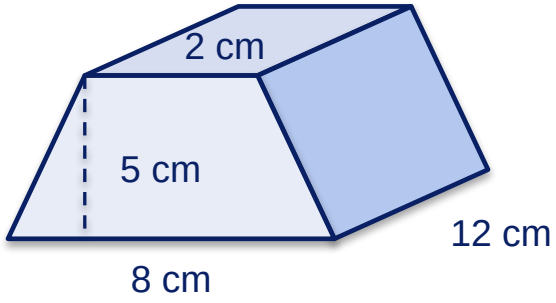
3 **210 cm³**



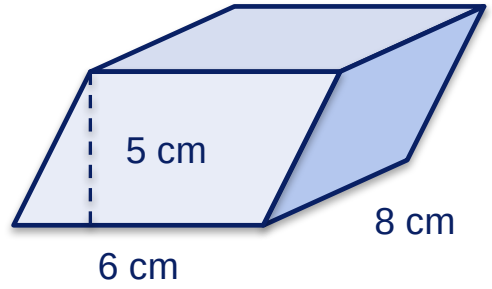
4 **180 cm³**



5 **300 cm³**



★ **30 × 8 = 240 cm³**



Examples 5 and 6 — Compound Prisms and Missing Lengths

EXAMPLES · 5 MIN

GOLD

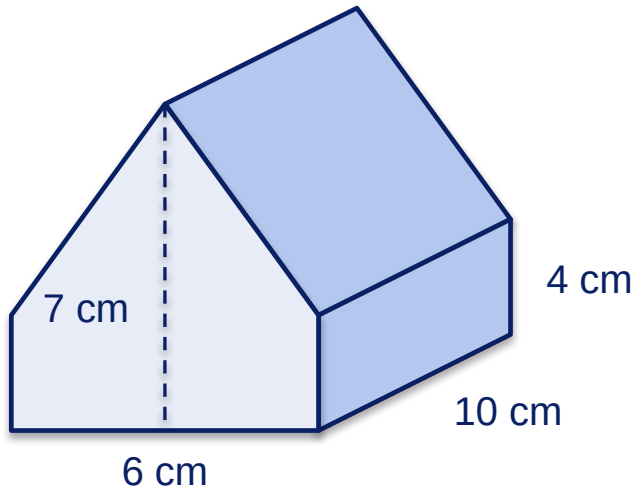
Exam-style: show all your working

Hint V = area of cross-section × length length = V ÷ area

Example 5

(3 marks)

Find the volume.



Example 6

(2 marks)

A triangular prism has volume 360 cm^3. The triangle has base 8 cm and height 6 cm.

Find the length of the prism.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Rectangle + triangle

$6 \times 4 = 24, \quad \frac{1}{2} \times 6 \times 3 = 9$

$\text{area} = 33\ \text{cm}^2$

$V = 33 \times 10 = 330\ \text{cm}^3$

Tip: the triangle is $7 - 4 = 3$ cm high.

Example 6

Length = volume ÷ area

$\text{area} = \frac{1}{2} \times 8 \times 6 = 24\ \text{cm}^2$

$360 \div 24$

$= 15$ cm

Tip: check: $24 \times 15 = 360$

Task 3

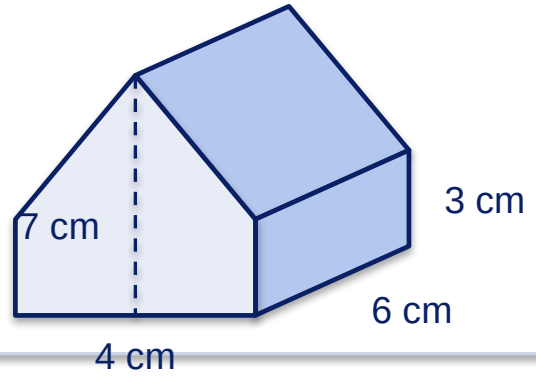
PRACTICE · 9 MIN

GOLD

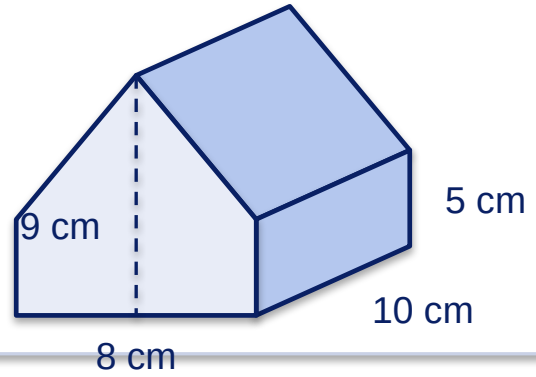
Find the volume, or the missing length

Hint V = area of cross-section × length
length = V ÷ area

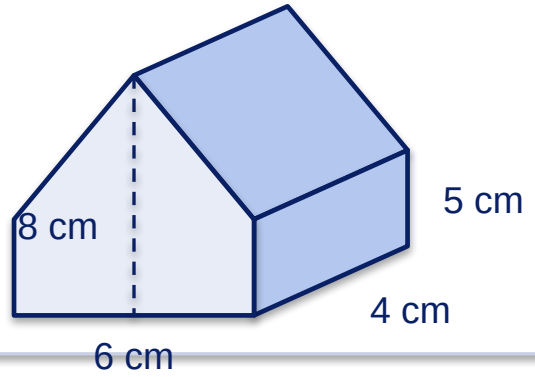
1 Volume



2 Volume



3 Volume



4 Find the height

Cuboid: V = 240 cm³, base 8 cm by 6 cm.

5 Find the length

Triangular prism: V = 300 cm³, triangle base 10 cm, height 6 cm.

★ Find the length

Trapezium prism: V = 420 cm³, parallel sides 4 cm and 10 cm, height 5 cm.

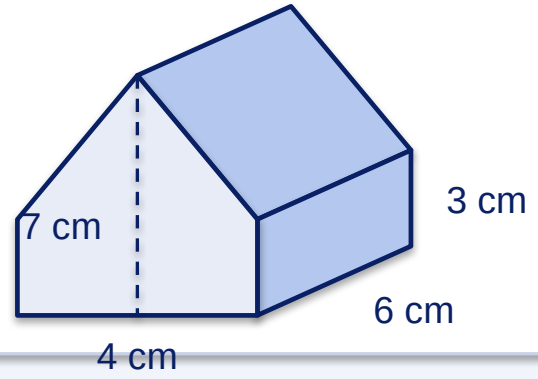
Task 3 — Answers

CORRECTIONS

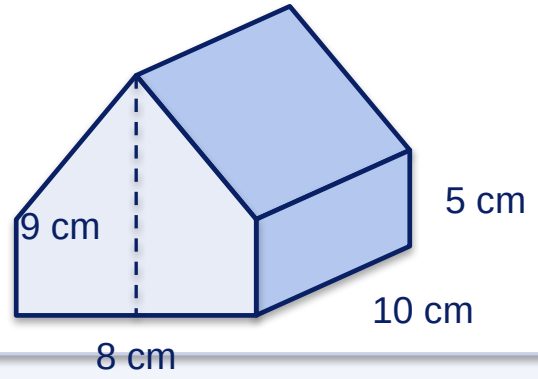
GOLD

Tick or correct in green pen

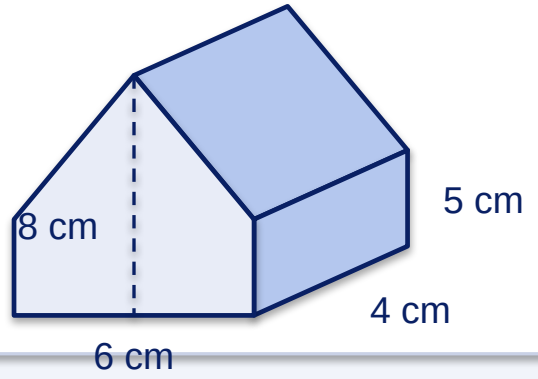
1 **120 cm³**



2 **560 cm³**



3 **156 cm³**



4 **5 cm**

base = 48 cm²

240 ÷ 48 = 5 cm

5 **10 cm**

area = 30 cm²

300 ÷ 30 = 10 cm

★ **12 cm**

area = 35 cm²

420 ÷ 35 = 12 cm

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Volume of a cuboid 4 cm by 5 cm by 6 cm

SILVER Trapezium prism: a = 2, b = 6, h = 3, length 10 (cm)

GOLD Volume 200 cm^3, cross-section 25 cm^2. Find the length.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **120 cm³**

SILVER **12 × 10 = 120 cm³**

GOLD **200 ÷ 25 = 8 cm**

Exam Practice — Volume

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

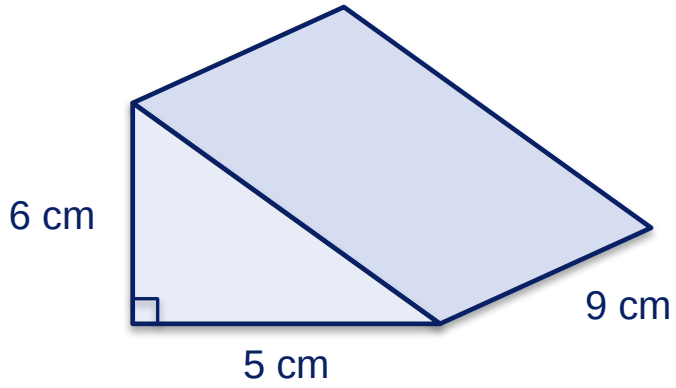
Hint V = area of cross-section × length
1000 cm³ = 1 litre

Question 1

(2 marks)

Work out the volume of this triangular prism.

Diagram NOT accurately drawn



..... cm³

Question 2

(3 marks)

A fish tank is 60 cm long, 30 cm wide and 40 cm high. How many litres of water does it hold when full?

..... litres

Exam Practice — Prisms

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

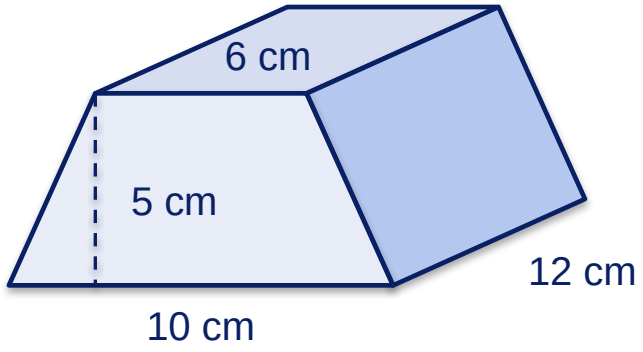
Hint trapezium = $\frac{1}{2}(a + b)h$

Question 3

(2 marks)

Work out the volume of this prism.

Diagram NOT accurately drawn



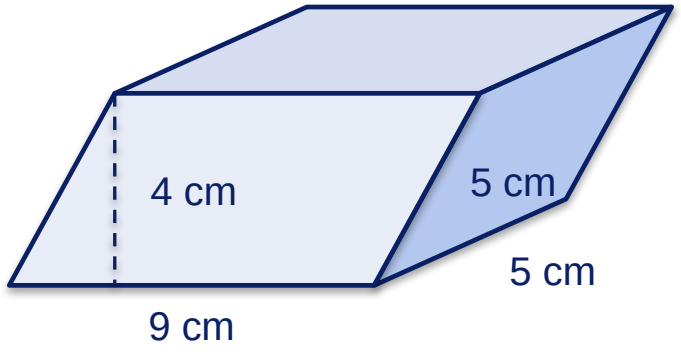
..... cm³

Question 4

(1 mark)

The diagram shows a prism.
Show that its volume is 180 cm³.

Diagram NOT accurately drawn



Exam Practice — Compound Prisms

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

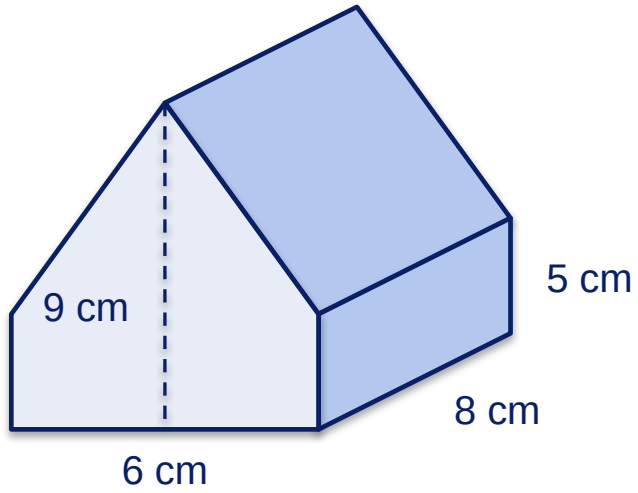
Hint V = area of cross-section × length
length = V ÷ area

Question 5

(3 marks)

Calculate the volume of this prism.

Diagram NOT accurately drawn



..... cm³

Question 6

(3 marks)

A prism has volume 600 cm³.
Its cross-section is a trapezium with parallel sides 7 cm and 13 cm and height 6 cm.
Find the length of the prism.

..... cm

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$\frac{1}{2} \times 5 \times 6 = 15$ — M1

$15 \times 9 = 135\ cm^3$ — A1

Tip: area of the cross-section × length

Question 2

$60 \times 30 \times 40$ — M1

$= 72\,000\ cm^3$ — A1

$\div 1000 = 72$ litres — A1

Tip: 1000 cm^3 = 1 litre

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$\frac{1}{2} \times (6 + 10) \times 5 = 40$ **M1**

$40 \times 12 = 480\ cm^3$ **A1**

Tip: the trapezium is the cross-section.

Question 4

$9 \times 4 \times 5 = 180\ cm^3$ **B1**

Tip: use the height 4 cm, not the slant 5 cm.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$6 \times 5 = 30, \quad \frac{1}{2} \times 6 \times 4 = 12$ — M1

42×8 — M1

$= 336\ \text{cm}^3$ — A1

Tip: the triangle is 9 − 5 = 4 cm high.

Question 6

$\frac{1}{2} \times (7 + 13) \times 6 = 60$ — M1

$600 \div 60$ — M1

$= 10$ cm — A1

Tip: length = volume ÷ area